Letter of Inquiry

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How can we reliably measure misinformation online?

The spread of misinformation has been studied extensively by scholars from various scientific disciplines, therefore, relying on diverse conceptualizations. Definitions of misinformation seem to largely depend on the respective methodological availabilities or choices, resulting in imprecise or inconclusive findings (Altay et al., 2023). To track misinformation on social media, computational researchers tend to use factchecked stories (Vosoughi et al., 2018), or follow blacklisted domains (González-Bailón et al., 2023; Guess et al., 2019; Grinberg et al., 2019; Robertson et al., 2023). While the applicability of story vs. source-level approaches depends on the specific research objective, the latter has been shown to be both scalable and reliable (Lin et al., 2023), yet suffering from low constuct validity—as unreliable domains do not equal misinformation. Lazer. Therefore, we review different approaches to measuring misinformation online in terms of reliability, validity and feasibility and then focus on one perspective in particular that has recently gained popularity: domain-based approaches.

Taken together, this proposal outlines research built on the question to what extent list-based approaches can reliably measure misinformation online (over time and across contexts).

In the United States, one of the most prominent lists used for assessing misinformation is curated by the organization NewsGuard. NewsGuard's analysts employ an expert-based approach; assigning ratings based on nine distinct criteria and regularly updating these assessments. However, there are several challenges associated with this list: First, the domain ratings were first created as a tool for brand security rather than a research tool, which is why the list is propriatory and access as well as publication of the data is restricted. Second, the coverage of the list has so far only been validated for research purposed for the U.S. context, where correlations were high given NewsGuards focus (Lin et al., 2023). Still, it is crucial to assess the usefulness of this list for research beyond the English-speaking world. Third, the list is updated in unknown intervals, which is why it is important to understand the update process and its implications for research. Fourth, the list is not exhaustive, which is why it is important to understand the selection process and its implications for research. Fifth, the list is not free of errors and missing data points, which is why it is important to understand the reliability of the ratings and their implications for research.

Consequently, we propose to provide a comprehensive description of this database over time (monthly granularity), outline its potential drawbacks for different contexts, and ultimately offer recommendations for its application. More precisely, we believe that is is essential to address the following aspects of the database:

a) the construction of the ratings,

b) the rating update process,

c) the selection and removal of sources, and

d) the reliability of alternative information, such as political orientation.

In addition, we reproduce published research relying on the NewsGuard list with different versions of the list to assess to what degree results depend on a specific version of the list.

Consequences of misinformation are manifold and may range from political polarization (Guess et al., 2023) to vaccine hesitancy (Brennen et al., 2023). Therefore, it is crucial to understand the spread of misinformation and its effects on society. In order to do so, we need to rely on reliable and valid data (Williams), especially when this research influences policy-making. However, when our data is biased because of an incomplete or case-specific selection of domains, so will be decisions and ultimately, digital media.